

The Natural Object

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Abstract

This paper constructs a finite coherent algebra $G^\natural$ from the extended binary Golay code and proves its closure as a finite object. The construction begins with three copies of the extended Hamming code $H_8 = \text{RM}(1, 3)$ inside $\mathbb{F}_2^{24}$. The direct sum has exactly forty-two weight-four roots. Three explicit Type II neighbor operations kill these roots in the sequence $42 \rightarrow 18 \rightarrow 6 \rightarrow 0$, and the endpoint is the extended binary Golay code G_{24} . The middle shell $G_{24}^{(12)}$ has 2576 dodecads. A lifted coorient group $\tilde{B}_3$ of order 9216 acts on this shell with six object orbits; its action on ordered pairs has 985 orbitals. The natural object is the corresponding orbital algebra

$$G^\natural = A_{985} = \text{End}_{\tilde{B}_3}(\mathbb{Q}[G_{24}^{(12)}]).$$

The algebra has a sparse multiplication tensor with 1,414,965 nonzero structure constants, center dimension 39, Wedderburn block sizes

$$1^7, 2^8, 3^4, 4^8, 5^4, 6^2, 8, 9, 10^2, 11, 12,$$

and coherent-configuration automorphism group exactly $\tilde{B}_3$. The quotient chain $A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}$ and the uniform six-object simple sector are derived as internal shadows of the same algebra. The paper is self-scoped: it proves the finite object and supplies a reproducibility certificate for each finite count.

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1 The object to be proved

We begin with the point at which the construction first becomes rigid. A single extended Hamming code H_8 has fourteen words of weight four. Three independent copies therefore carry forty-two such words. They are the only roots present at the source stage. The paper proves that there is a finite passage

$$H_8^{\oplus 3} \longrightarrow G_{24}$$

which removes exactly those forty-two roots, preserves the Type II condition, and lands at the extended binary Golay code. From that point the construction is no longer a code-only story. The dodecad shell of G_{24} has a fixed sextet geometry; the code-preserving coorient action on this shell closes into a 985-orbital coherent configuration; and the algebra of that coherent configuration is $G^\natural$.

The proof is organized around one finite diagram:

$$H_8^{\oplus 3} \xrightarrow{42 \rightarrow 18 \rightarrow 6 \rightarrow 0} G_{24} \xrightarrow{\text{wt} = 12} X = G_{24}^{(12)} \xrightarrow{\text{lift}} \tilde{B}_3 \curvearrowright X \xrightarrow{\text{orbitals}} X \times X / \tilde{B}_3 \xrightarrow{\text{incidence}} A_{985}$$

Every arrow is finite. The first is a neighbor calculation in binary coding theory. The second is extraction of a weight shell. The third is a finite group action. The fourth is an orbit calculation on ordered pairs. The fifth is multiplication by two-step incidence. Closure means that after this chain is completed, the object has no unaccounted symmetry, no missing simple block, and no unverified quotient.

1.1 Main theorem

Theorem 1.1 (The Natural Object). *Let $H_8 = \text{RM}(1, 3)$ and let $C_0 = H_8^{\oplus 3} \subset \mathbb{F}_2^{24}$. Let C_3 be obtained from C_0 by the three Type II neighbor operations along*

$$\begin{aligned} v_1 &= \{2, 4, 9, 10, 12, 13, 14, 16, 17, 18, 20, 22\}, \\ v_2 &= \{3, 4, 9, 13, 15, 16, 19, 20\}, \\ v_3 &= \{2, 5, 7, 8, 9, 10, 12, 14, 15, 16, 17, 18, 19, 21, 22, 23\}. \end{aligned}$$

Then $C_3 = G_{24}$. If $X = G_{24}^{(12)}$ and $\tilde{B}_3$ is the lifted code-preserving coorient group attached to the fixed sextet, then

$$G^\natural = A_{985} = \text{End}_{\tilde{B}_3}(\mathbb{Q}[X])$$

is a semisimple coherent algebra of dimension 985 with multiplication tensor having 1,414,965 nonzero entries. Its center has dimension 39, its Wedderburn block sizes are

$$1^7, 2^8, 3^4, 4^8, 5^4, 6^2, 8, 9, 10^2, 11, 12,$$

and

$$\text{Aut}(G^\natural) = \tilde{B}_3, \quad |\tilde{B}_3| = 9216.$$

The quotients $A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}$ close, and the simple sector with id 20 is a uniform six-object bridge with degree 6 and multiplicity vector $(1, 1, 1, 1, 1, 1)$.

The theorem is proved in stages. The proof does not require the reader to trust a table of constants. The finite counts are obtained from the definitions, then checked by certificates. The role of the certificates is the same as the role of a determinant expansion in an ordinary finite proof: they make a large but elementary finite calculation reproducible.

2 The Hamming source

Let $\mathbb{F}_2^3$ be the affine three-space over $\mathbb{F}_2$. Index the eight coordinates of one lane by points $x \in \mathbb{F}_2^3$. The extended Hamming code is

$$H_8 = \{x \mapsto a_0 + a \cdot x : a_0 \in \mathbb{F}_2, a \in \mathbb{F}_2^3\}.$$

It has dimension 4. The constant function 1 has weight 8, the nonconstant affine functions have weight 4, and their complements also have weight 4. Hence

$$W_{H_8}(y) = 1 + 14y^4 + y^8.$$

Place three such lanes on

$$\{1, \dots, 8\}, \quad \{9, \dots, 16\}, \quad \{17, \dots, 24\}.$$

The source code is

$$C_0 = H_8 \oplus H_8 \oplus H_8.$$

The weight enumerator is therefore

$$\begin{aligned} W_{C_0}(y) &= (1 + 14y^4 + y^8)^3 \\ &= 1 + 42y^4 + 591y^8 + 2828y^{12} + 591y^{16} + 42y^{20} + y^{24}. \end{aligned}$$

The coefficient 42 is the first obstruction. It is literally the set of weight-four roots of C_0 .

2.1 The neighbor operation

For a binary code $C \subset \mathbb{F}_2^{24}$ and a vector $v \notin C$ with $\text{wt}(v) \equiv 0 \pmod{4}$, define

$$N_v(C) = \langle C \cap v^\perp, v \rangle.$$

If C is Type II, then $N_v(C)$ is again Type II. Orthogonality cuts one hyperplane from C and adjoining v restores the dimension. Doubly-evenness follows from the weight condition and from v being orthogonal to $C \cap v^\perp$.

Set

$$C_i = N_{v_i}(C_{i-1}), \quad i = 1, 2, 3.$$

The finite calculation is short enough to state without ornament:

$$\#\{c \in C_i : \text{wt}(c) = 4\} = 42, 18, 6, 0 \quad \text{for } i = 0, 1, 2, 3.$$

The endpoint C_3 is Type II of length 24, dimension 12, and minimum weight 8.

Proposition 2.1 (root killing). *The three displayed neighbor operations take $C_0 = H_8^{\oplus 3}$ to G_{24} .*

Proof. Each step preserves the Type II class. The displayed root count shows that C_3 has no weight-four word. A Type II code of length 24 and minimum weight 8 is the extended binary Golay code; in the certificate this recognition is performed concretely by matching the generated code with the standard Golay generator used for dodecad enumeration. Thus $C_3 = G_{24}$ up to the fixed coordinate convention. $\square$

Root-killing sheet.

$$\begin{aligned} H_8^{\oplus 3} & : 42 \text{ roots,} \\ N_{v_1}(H_8^{\oplus 3}) & : 18 \text{ roots,} \\ N_{v_2}N_{v_1}(H_8^{\oplus 3}) & : 6 \text{ roots,} \\ N_{v_3}N_{v_2}N_{v_1}(H_8^{\oplus 3}) & : 0 \text{ roots.} \end{aligned}$$

The zero-root endpoint has enumerator

$$1 + 759y^8 + 2576y^{12} + 759y^{16} + y^{24}.$$

3 The dodecad shell and the sextet

Let

$$X = G_{24}^{(12)} = \{x \in G_{24} : \text{wt}(x) = 12\}.$$

Then $|X| = 2576$. The elements of X are dodecads.

Fix the sextet of tetrads

$$\begin{aligned} T_0 &= \{0, 1, 2, 3\}, & T_1 &= \{4, 5, 6, 7\}, & T_2 &= \{8, 9, 12, 13\}, \\ T_3 &= \{10, 11, 14, 15\}, & T_4 &= \{16, 17, 22, 23\}, & T_5 &= \{18, 19, 20, 21\}. \end{aligned}$$

For a dodecad D , write

$$p(D) = (|D \cap T_0|, \dots, |D \cap T_5|).$$

Since D has twelve coordinates, the entries of $p(D)$ sum to 12. The Golay condition forces only three profile types:

$$(2^6), \quad (4, 0, 2^4), \quad (3^3, 1^3).$$

Their cardinalities are

$$576, \quad 720, \quad 1280.$$

Thus

$$2576 = 576 + 720 + 1280.$$

This first split should not be confused with the final six-object split. It is the visible profile geometry from which the lifted coorient action later refines the shell. The balanced part (2^6) is the rigid core. The vector part $(4, 0, 2^4)$ marks a full tetrad and an empty tetrad. The spinor-state part $(3^3, 1^3)$ marks a choice of three heavy and three light tetrads.

3.1 The balanced scheme

In the balanced stratum every tetrad contributes two chosen coordinates. For two balanced dodecads D, E , the ordinary Hamming distance is less informative than the column profile of their difference. Define

$$n_j(D, E) = \#\{i : |D \cap T_i \triangle E \cap T_i| = j\}, \quad j = 0, 2, 4.$$

The relation type is

$$\rho(D, E) = (n_0, n_2, n_4).$$

Exactly eight types occur:

$$\begin{aligned} (6, 0, 0), (2, 4, 0), (4, 0, 2), (0, 6, 0), \\ (1, 4, 1), (0, 4, 2), (2, 0, 4), (0, 0, 6). \end{aligned}$$

The corresponding valencies are

$$1, 120, 15, 64, 240, 120, 15, 1.$$

This is an association scheme on 576 points. After the identity relation is removed, it is a seven-class symmetric association scheme. The balanced scheme is the first internal algebraic shadow, but not the natural object itself. The natural object appears only after the entire dodecad shell is acted on by the lifted group.

3.2 Vector and spinor-state fibers

The vector profile $(4, 0, 2^4)$ has a distinguished full tetrad and a distinguished empty tetrad. The ordered choice of those opposite roles gives thirty fibers of size 24. The spinor-state profile $(3^3, 1^3)$ has twenty affine $[12, 6]$ fibers of size 64. These two fiber laws are used in the certificate as local checks on the dodecad enumeration. They also explain why the final six object labels contain both vector and spinor-state names.

4 The lifted coorient group

The sextet comes with three opposite pairs of tetrads:

$$(T_0, T_3), \quad (T_1, T_4), \quad (T_2, T_5).$$

The visible signed-pair grammar is

$$C_2^3 \rtimes S_3 = C_2 \wr S_3.$$

Inside each tetrad there are four coordinates, so the ambient imprimitive group sits inside

$$S_4^6 \rtimes (C_2 \wr S_3).$$

The Golay code selects the subgroup of this ambient group that preserves the terminal code G_{24} .

Definition 4.1. The lifted coorient group $\tilde{B}_3$ is the subgroup of the sextet imprimitive group that preserves G_{24} and the fixed coorient data of the sextet.

Calculation 4.2 (coorient enumeration). The code-preserving enumeration gives

$$|\tilde{B}_3| = 9216.$$

The action of $\tilde{B}_3$ on $X = G_{24}^{(12)}$ has six orbits with sizes

$$384, 192, 144, 576, 512, 768.$$

The six orbits are denoted

$$B_{384}, \quad B_{192}, \quad V_6, \quad V_{24}, \quad S_8, \quad S_{12}.$$

The subscripts are names of the six fields of the object, not always their orbit sizes. The actual sizes in this order are

$$384, 192, 144, 576, 512, 768.$$

This fixed order will be used for block matrices.

5 Orbitals and coherent configurations

Let $G = \tilde{B}_3$ and let $X = G_{24}^{(12)}$. The permutation representation $\mathbb{Q}[X]$ decomposes according to the six G -orbits on X , but the algebra of G -equivariant endomorphisms is finer. It is controlled by orbits on ordered pairs.

For $x, y \in X$, write

$$(x, y) \sim (x', y') \iff (x', y') = (gx, gy) \text{ for some } g \in G.$$

The equivalence classes are orbitals $R_\alpha \subset X \times X$. Let A_α be the adjacency matrix of R_α . Then

$$A = \text{span}_{\mathbb{Q}}\{A_\alpha\}$$

is a coherent algebra. Matrix multiplication is determined by intersection numbers

$$p_{\alpha\beta}^\gamma = \#\{z \in X : (x, z) \in R_\alpha, (z, y) \in R_\beta\}, \quad (x, y) \in R_\gamma.$$

The number is independent of the representative (x, y) because R_γ is a G -orbit.

Lemma 5.1. *The orbital span is canonically*

$$\text{End}_G(\mathbb{Q}[X]).$$

Proof. A G -equivariant endomorphism of $\mathbb{Q}[X]$ is a matrix whose entries are constant on G -orbits in $X \times X$. Such matrices are exactly the $\mathbb{Q}$ -linear combinations of orbital adjacency matrices. Hence the orbital span equals the commutant of the permutation representation. $\square$

Definition 5.2. The natural object is

$$G^\natural = A_{985} = \text{End}_{\tilde{B}_3}(\mathbb{Q}[X]).$$

Calculation 5.3 (the pair closure). The action of $\tilde{B}_3$ on $X \times X$ has 985 orbitals. Therefore

$$\dim G^\natural = 985.$$

The sparse tensor $p_{\alpha\beta}^\gamma$ has 1,414,965 nonzero entries.

The six object-orbit decomposition gives a visible block matrix recording how many orbitals run from one object orbit to another:

$$M = \begin{pmatrix} 33 & 12 & 8 & 24 & 30 & 32 \\ 12 & 16 & 4 & 12 & 12 & 20 \\ 8 & 4 & 14 & 22 & 8 & 20 \\ 24 & 12 & 22 & 86 & 32 & 64 \\ 30 & 12 & 8 & 32 & 44 & 44 \\ 32 & 20 & 20 & 64 & 44 & 104 \end{pmatrix}.$$

Its entries sum to 985. This matrix is the Mackey shadow of the full coherent algebra. It is useful because it records the six-object field of the algebra, but it is not a replacement for the 985-relation tensor.

6 Semisimplicity and Wedderburn closure

Since G is finite and the ground field is $\mathbb{Q}$, the group algebra $\mathbb{Q}[G]$ is semisimple. The permutation module $\mathbb{Q}[X]$ is therefore semisimple, and the commutant

$$A_{985} = \text{End}_G(\mathbb{Q}[X])$$

is semisimple. Hence

$$A_{985} \cong \bigoplus_{\lambda} \text{Mat}_{m_{\lambda}}(D_{\lambda})$$

for finite division algebras D_{λ} over number fields. The certificate works over split finite primes where all center factors split linearly, recovering the matrix sizes m_{λ} .

The center is the solution space of the commutation equations

$$zA_{\alpha} = A_{\alpha}z, \quad \alpha = 0, \dots, 984.$$

After diagonal reductions there are 297 unknowns. Over the checked split primes the center has dimension 39.

Proposition 6.1. *The Wedderburn block sizes of $G^{\natural}$ are*

$$1^7, 2^8, 3^4, 4^8, 5^4, 6^2, 8, 9, 10^2, 11, 12.$$

The sum of their squares is 985.

Proof. The certificate computes a basis of the center, chooses a generic central element, and factors its minimal polynomial over the checked split primes. Each linear factor gives a primitive central idempotent e_{λ} . The dimension of the corresponding simple summand is $m_{\lambda}^2 = \dim(Ae_{\lambda})$. The resulting sizes are those displayed. Their squares add to 985, so all of A_{985} is accounted for. The identity diagnostic verifies that the geometric diagonal identity acts as a left and right identity. $\square$

The center dimension is therefore not merely a rank measurement. It is the number of simple sectors. These sectors are the irreducible pieces through which the six object orbits communicate.

7 Object multiplicities and the uniform sector

Let

$$1 = e_1 + \dots + e_6$$

be the sum of the six object idempotents corresponding to the six G -orbits in X . If E_{λ} is a primitive central idempotent, then each $e_i E_{\lambda}$ has some matrix rank $d_{\lambda,i}$. Thus each simple sector has a multiplicity vector

$$d_{\lambda} = (d_{\lambda,1}, \dots, d_{\lambda,6}).$$

These vectors reconstruct the six-object Mackey matrix by

$$M_{ij} = \sum_{\lambda} d_{\lambda,i} d_{\lambda,j}.$$

This identity is the finite reason the six-object structure is part of the algebra rather than a decoration placed afterward.

Proposition 7.1 (uniform bridge). *The simple sector with computational id 20 has degree*

$$m = 6$$

and multiplicity vector

$$d = (1, 1, 1, 1, 1, 1).$$

Its object contribution is the 6×6 all-ones matrix.

Proof. The matrix model atlas constructs explicit matrix units for every simple sector. In sector 20, the object idempotents restrict to six rank-one diagonal projections in Mat_6 . Therefore every object orbit occurs exactly once. Since the matrix units span the full Mat_6 , the sector is simple and has degree 6. $\square$

This is the cleanest internal sign that the algebra has a genuine six-object bridge. The sector is not selected at the beginning. It appears only after the full 985-orbital algebra has been decomposed.

8 The quotient tower

The natural object has three smaller closed shadows:

$$A_{985} \longrightarrow A_{236} \longrightarrow A_{42} \longrightarrow A_{12}.$$

The first quotient is obtained by fusing orbitals according to support data. Starting from the initial support partition, repeated closure under multiplication gives

$$191 \longrightarrow 228 \longrightarrow 234 \longrightarrow 236.$$

At 236 the partition is stable, giving the algebra A_{236} . Its tensor has 37,787 nonzero entries, its center has dimension 34, and its block sizes are

$$1^{20}, 3^{10}, 4, 5, 6, 7.$$

The sum of squares is

$$20 \cdot 1^2 + 10 \cdot 3^2 + 4^2 + 5^2 + 6^2 + 7^2 = 236.$$

The uniform sector behaves well under this quotient. The full sector 20 descends as a single 6-dimensional component in A_{236} . At the 42-level it is the component denoted $T6$. Under the vector-rule restriction to A_{12} it becomes

$$T6 \downarrow A_{12} = 2T3.$$

Thus the six-object bridge survives the first two reductions and splits only at the final 12-class shadow.

9 Automorphism closure

The action group $\tilde{B}_3$ acts by color-preserving automorphisms of the coherent configuration. Hence

$$\tilde{B}_3 \leq \text{Aut}(G^\natural).$$

The issue is whether a larger color-preserving group exists after the orbitals have been formed.

The certificate uses a finite base. Choose dodecads $b_1, \dots, b_r$ and define the two-sided orbital fingerprint

$$\Phi_B(x) = (\text{orb}(b_1, x), \dots, \text{orb}(b_r, x), \text{orb}(x, b_1), \dots, \text{orb}(x, b_r)).$$

The base is selected so that Φ_B is injective. A coherent-configuration automorphism is then determined by the image of the base. The algorithm enumerates every image of the base compatible with all orbital colors. It finds exactly 9216 possibilities, each realized by an element of $\tilde{B}_3$.

Theorem 9.1 (automorphism closure). *The coherent-configuration automorphism group is*

$$\text{Aut}(G^\natural) = \tilde{B}_3.$$

In particular its order is 9216.

Proof. The inclusion $\tilde{B}_3 \leq \text{Aut}(G^\natural)$ follows from construction. For the reverse inclusion, the separating base makes any automorphism determined by a finite base image. The color-compatible base-image search gives precisely the 9216 images realized by $\tilde{B}_3$. Hence no additional automorphism exists. $\square$

10 The calculation as a derivation

A finite calculation may enter a proof in two different ways. It may be a table of values attached to a construction already understood, or it may be the construction itself. Here it is the latter. The algebra $G^\natural$ is not first defined abstractly and then measured; it is defined by the finite operations that produce its basis, tensor, center, and automorphism group. This section records the derivation in a form that can be checked without reading the source code line by line.

10.1 The three constraints that force the object

Three constraints determine the construction.

First, the source must be Type II. This is why $H_8^{\oplus 3}$ is the correct starting point: each Hamming lane is already doubly even and self-dual, and the direct sum keeps both properties. The forty-two roots are therefore not a defect of parity or dimension. They are the only remaining obstruction to being the Golay endpoint.

Second, the passage must remain inside the Type II neighbor graph. An arbitrary vector could kill roots, but it would not preserve the structure needed for a length-24 Type II endpoint. The neighbor formula

$$N_v(C) = \langle C \cap v^\perp, v \rangle$$

keeps the code inside the correct component of the self-dual landscape whenever v is admissible. The vectors v_1, v_2, v_3 are therefore not corrections after the fact; they are the three chosen walls across which the Type II code is moved.

Third, the endpoint must be read through dodecads. The full code has 4096 words, but the natural finite geometry lives in the 2576 words of weight 12. This is the unique shell large enough to display the sextet profile split, the vector fibers, the spinor-state fibers, and the six lifted object orbits simultaneously. The 985-orbital algebra is a closure of this shell, not of the vector space $\mathbb{F}_2^{24}$.

Forcing diagram.

$$\text{Type II source} \xrightarrow{42 \text{ roots}} \text{Type II neighbors} \xrightarrow{0 \text{ roots}} G_{24} \xrightarrow{2576} G_{24}^{(12)} \xrightarrow{985} A_{985}$$

The numbers are not independent invariants. They are successive residues of the same finite path.

10.2 Why the root count is the right first invariant

The code C_0 is already self-dual and doubly even. Therefore most of the usual obstructions to the Golay code have already vanished. What remains is the minimum weight. The extended Golay code is the Type II length-24 code with no roots. Thus the root count is the smallest possible invariant detecting failure of the desired endpoint.

Let

$$R(C) = \{c \in C : \text{wt}(c) = 4\}.$$

Then $R(C_0)$ consists of three disjoint copies of the 14 affine-plane words of H_8 . Since a neighbor move cuts by $v^\perp$ and adjoins v , it changes the root set in two ways:

$$R(C) \cap v^\perp \quad \text{survives directly,}$$

and roots not in the hyperplane may reappear only through sums with the new vector v . The certificate tests both possibilities exactly. The fact that the count becomes

$$42, 18, 6, 0$$

means that the chosen neighbors do not merely reduce the number of roots; they remove the root system in three controlled collapses. At zero roots the endpoint is forced.

10.3 Why the dodecad shell is not optional

After the Golay code is obtained, there are several finite sets one might choose: octads, dodecads, complements, coordinate positions, or all codewords. The natural object uses dodecads because the sextet profile becomes a balanced three-lane geometry exactly at weight 12. In a dodecad, the six tetrad intersection sizes sum to 12, and the allowed profiles

$$(2^6), \quad (4, 0, 2^4), \quad (3^3, 1^3)$$

are exactly the balanced, vector, and spinor-state profiles. The three numbers

$$576, \quad 720, \quad 1280$$

are therefore not arbitrary orbit sizes. They are the three ways a dodecad can occupy the sextet.

The lifted coorient action refines this profile data. It does not erase it. The six lifted object orbits are the stable orbit-level reading of the same dodecad shell under the code-preserving coorient group. This is why the algebra begins with six object idempotents, even though its full relation basis has 985 elements.

11 The Mackey reading of the six-object field

Let the six object orbits be $X_1, \dots, X_6$. Then

$$\mathbb{Q}[X] = \bigoplus_{i=1}^6 \mathbb{Q}[X_i].$$

The algebra decomposes into blocks

$$e_i A_{985} e_j = \text{Hom}_G(\mathbb{Q}[X_j], \mathbb{Q}[X_i]),$$

where e_i is the projection to $\mathbb{Q}[X_i]$. Consequently

$$M_{ij} = \dim e_i A_{985} e_j$$

is a Mackey-type matrix. The displayed matrix M is therefore not a table of unrelated counts; it is the dimension matrix of all equivariant passages between the six object fields.

This gives the conceptual role of the matrix

$$M = \begin{pmatrix} 33 & 12 & 8 & 24 & 30 & 32 \\ 12 & 16 & 4 & 12 & 12 & 20 \\ 8 & 4 & 14 & 22 & 8 & 20 \\ 24 & 12 & 22 & 86 & 32 & 64 \\ 30 & 12 & 8 & 32 & 44 & 44 \\ 32 & 20 & 20 & 64 & 44 & 104 \end{pmatrix}.$$

The row sums

$$139, 76, 76, 240, 170, 284$$

measure how many relation colors leave each object. The total sum is

$$139 + 76 + 76 + 240 + 170 + 284 = 985.$$

Thus the six-object field and the 985-relation field are the same algebra seen at two resolutions.

Lemma 11.1. *The matrix M is recovered from the simple-sector multiplicity vectors by*

$$M_{ij} = \sum_{\lambda} d_{\lambda,i} d_{\lambda,j}.$$

Proof. In the semisimple algebra $A = \oplus_{\lambda} \text{Mat}_{m_{\lambda}}$, the idempotent e_i restricts in the λ th block to a projection of rank $d_{\lambda,i}$. Therefore the space $e_i \text{Mat}_{m_{\lambda}} e_j$ has dimension $d_{\lambda,i} d_{\lambda,j}$. Summing over λ gives $\dim e_i A e_j = M_{ij}$. $\square$

This formula is more important than the printed matrix. It says the object-block relation counts are exactly the Gram matrix of the simple-sector rank vectors. The uniform sector 20 contributes the all-ones Gram summand because its vector is $(1, 1, 1, 1, 1, 1)$.

12 The algebra behind the tensor count

The number 1,414,965 is the size of the support of the multiplication tensor. It is large, but its definition is not complicated. For each orbital R_{γ} , choose a representative pair (x, y) . The product $A_{\alpha} A_{\beta}$ at the entry (x, y) counts intermediate points z with

$$(x, z) \in R_{\alpha}, \quad (z, y) \in R_{\beta}.$$

Thus

$$p_{\alpha\beta}^{\gamma} = \#\{z : (x, z) \in R_{\alpha}, (z, y) \in R_{\beta}\}.$$

The tensor support is

$$\{(\alpha, \beta, \gamma) : p_{\alpha\beta}^\gamma \neq 0\}.$$

The certificate finds its cardinality to be 1,414,965.

The reason this count matters is closure. If a proposed fusion or quotient of the orbitals is to define a smaller algebra, it must be stable under all these two-step counts. The quotient A_{236} is obtained precisely by enforcing that stability. Therefore the tensor is the object that governs both the full algebra and its finite shadows.

Proposition 12.1. *Every quotient algebra appearing in the chain*

$$A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}$$

is a fusion of the coherent configuration relations.

Proof. A fusion is a partition of the orbital basis. It defines a quotient relation algebra precisely when products of sums of classes are again sums of classes with well-defined structure constants. The closure procedure starts with the relevant support partition and repeatedly refines it until all products are stable. The fixed points of that procedure are the quoted algebras. Thus every arrow in the tower is a finite fusion quotient. $\square$

13 A direct proof of semisimple closure

The representation-theoretic closure can be written without referring to the computed sector list. Let

$$\mathbb{Q}[X] \cong \bigoplus_{\rho \in \text{Irr}(G)} m_\rho V_\rho$$

be the decomposition of the permutation representation. Then the commutant is

$$\text{End}_G(\mathbb{Q}[X]) \cong \bigoplus_{\rho \in \text{Irr}(G)} \text{Mat}_{m_\rho}(\text{End}_G(V_\rho)).$$

Over a splitting field, $\text{End}_G(V_\rho)$ becomes the ground field, and the block degrees are the multiplicities m_ρ . The checked primes are chosen so the recovered center factors split. The list of degrees in the theorem is therefore the list of multiplicities with which irreducible constituents occur in $\mathbb{Q}[X]$ after scalar extension.

The equality

$$\sum_{\rho} m_\rho^2 = 985$$

has two readings. On the left it is representation theory. On the right it is the number of orbitals. The equality is the double-centralizer identity for the action. It is also a check that no simple summand is missing.

14 The role of packet 20

The uniform sector is not selected because of its label. The id 20 is only the computational name of the primitive central idempotent in the atlas. Its intrinsic content is

$$m = 6, \quad d = (1, 1, 1, 1, 1, 1).$$

This means that in the corresponding simple block Mat_6 , the six object idempotents cut out six one-dimensional coordinate axes. Each object participates exactly once. The sector therefore supplies a canonical six-way bridge inside the algebra.

Under the quotient to A_{236} , this sector remains one 6-dimensional block. Under the 42-class quotient it is the $T6$ sector. Under the 12-class vector-rule restriction it splits as

$$T6 \mapsto 2T3.$$

The split at the last stage is not a failure. It records the fact that the 12-class shadow can no longer see all six object directions independently; it sees them as two ternary halves.

15 The automorphism proof in finite terms

A coherent configuration can have automorphisms that are not visible in the group used to construct it. This is why the equality

$$\text{Aut}(G^\natural) = \tilde{B}_3$$

is a theorem rather than a restatement.

The base-fingerprint proof is a finite version of rigidifying a relational structure by enough marked probes. The base $B = (b_1, \dots, b_r)$ is chosen so that every dodecad x is uniquely identified by the colors of the arrows to and from the base. If a color-preserving automorphism fixes the image of the base, it fixes every fingerprint and hence every point. Therefore every automorphism is determined by a valid base image.

The search then becomes finite and exact:

$$\{\text{extcompatiblebaseimages}\} = \{\text{extbaseimagesinducedby}\tilde{B}_3\}.$$

Both sets have size 9216. Hence the constructed group is the full group.

16 Normal form of the final theorem

The finite theorem can be compressed into four equations:

$$\begin{aligned} H_8^{\oplus 3} &\xrightarrow{N_{v_1}, N_{v_2}, N_{v_3}} G_{24}, \\ X &= G_{24}^{(12)}, \quad |X| = 2576, \\ G^\natural &= \text{End}_{\tilde{B}_3}(\mathbb{Q}[X]), \quad \dim G^\natural = 985, \\ \text{Aut}(G^\natural) &= \tilde{B}_3, \quad |\tilde{B}_3| = 9216. \end{aligned}$$

Everything else in the paper is the verification that these four equations are not slogans: the root-killing vectors are explicit, the dodecad shell is enumerated, the orbitals multiply coherently, the Wedderburn sectors exhaust the algebra, the quotient tower closes, and the automorphism group has no enlargement.

17 Referee-level reproducibility

A referee need not inspect all 1,414,965 tensor entries. The proof is reproducible by independent reruns of the following finite checks.

- (1) Rebuild H_8 from affine linear functions on $\mathbb{F}_2^3$.
- (2) Form $H_8^{\oplus 3}$ and verify the enumerator coefficient 42 at y^4 .
- (3) Apply the three neighbor moves and check the root sequence 42, 18, 6, 0.
- (4) Enumerate the weight-12 shell of the terminal code and verify cardinality 2576.
- (5) Apply the sextet coorient maps and keep exactly the code-preserving subgroup of order 9216.
- (6) Compute the orbitals on ordered pairs and verify that there are 985.
- (7) Reconstruct the multiplication tensor by the incidence formula and verify the stored support hash.
- (8) Solve the center equations over the split primes and verify dimension 39 and block-size square sum 985.
- (9) Construct the matrix models and verify that the sector 20 idempotent cuts the six object idempotents in ranks (1, 1, 1, 1, 1, 1).
- (10) Run the base-fingerprint automorphism check and verify that the full color-preserving group has order 9216.

These checks are independent enough to catch the relevant mistakes. The root-killing step checks the code. The dodecad count checks the shell. The orbital count checks the action. The tensor count checks multiplication. The center and matrix models check representation theory. The automorphism certificate checks that the relation structure has no extra symmetry.

18 Why this closes the finite object

The object closes because the same finite data answer every structural question raised by the construction:

$$\begin{aligned}
\text{source obstruction} &= 42 \text{ roots in } H_8^{\oplus 3}, \\
\text{root removal} &= 42 \rightarrow 18 \rightarrow 6 \rightarrow 0, \\
\text{endpoint code} &= G_{24}, \\
\text{point set} &= G_{24}^{(12)}, |X| = 2576, \\
\text{acting group} &= \tilde{B}_3, |\tilde{B}_3| = 9216, \\
\text{relations} &= 985 \text{ orbitals on } X \times X, \\
\text{algebra} &= \text{End}_{\tilde{B}_3}(\mathbb{Q}[X]), \\
\text{simple sectors} &= 39, \\
\text{full symmetry} &= \text{Aut}(G^\natural) = \tilde{B}_3.
\end{aligned}$$

Nothing remains to be chosen after the sextet and the three neighbor vectors are fixed. The tensor is forced by incidence. The center is forced by commutation. The simple sectors are forced by central idempotents. The automorphism group is forced by the orbital colors. This is the finite closure claimed by the title.

A A single reconstruction algorithm

The proof can be condensed into a single finite reconstruction algorithm. The implementation used in the certificate is optimized, but the following version gives the mathematical shape of the whole computation. Its inputs are the three neighbor vectors and the sextet. Its outputs are the code endpoint, the dodecad shell, the acting group, the orbital colors, the tensor support, the center dimension, the block list, and the automorphism order.

```
# Mathematical reconstruction of  $G^{\text{natural}}$ .
# This is a proof kernel, not a speed-optimized implementation.

# Stage 0: binary linear algebra
# A binary vector is a tuple in  $\{0,1\}^n$ .

def add(u,v):
    return tuple(a ^ b for a,b in zip(u,v))

def dot(u,v):
    return sum(a*b for a,b in zip(u,v)) % 2

def wt(u):
    return sum(u)

def span(gens):
    if not gens:
        return []
    n=len(gens[0])
    S={(0,)*n}
    for g in gens:
        S |= {add(x,g) for x in list(S)}
    return sorted(S)

# Stage 1:  $H_8$  and  $H_8^3$ 
pts=[(a,b,c) for a in (0,1) for b in (0,1) for c in (0,1)]
Hgens=[tuple(1 for _ in pts)]
for j in range(3):
    Hgens.append(tuple(p[j] for p in pts))
H8=span(Hgens)
C0=[a+b+c for a in H8 for b in H8 for c in H8]

# Stage 2: Type II neighbors

def vec_from_one_based(S):
    return tuple(1 if i+1 in S else 0 for i in range(24))

def neighbor(C,v):
    return span([c for c in C if dot(c,v)==0] + [v])

v1=vec_from_one_based({2,4,9,10,12,13,14,16,17,18,20,22})
v2=vec_from_one_based({3,4,9,13,15,16,19,20})
v3=vec_from_one_based({2,5,7,8,9,10,12,14,15,16,17,18,19,21,22,23})
C=C0
root_history=[]
for v in (v1,v2,v3):
    root_history.append(sum(1 for c in C if wt(c)==4))
    C=neighbor(C,v)
root_history.append(sum(1 for c in C if wt(c)==4))
assert root_history == [42,18,6,0]
G24=C

# Stage 3: dodecad shell
X=[c for c in G24 if wt(c)==12]
assert len(X)==2576

# Stage 4: sextet profiles
sextet=[
```



```

    {0,1,2,3}, {4,5,6,7}, {8,9,12,13},
    {10,11,14,15}, {16,17,22,23}, {18,19,20,21}
]

def profile(D):
    S={i for i,b in enumerate(D) if b}
    return tuple(len(S & T) for T in sextet)

profiles={}
for D in X:
    profiles[profile(D)] = profiles.get(profile(D),0)+1
# The three profile families are (2^6), (4,0,2^4), and (3^3,1^3).

# Stage 5: lifted coorient group
# Enumerate the ambient sextet-imprimitive permutations, keep those preserving G24.
# In the optimized certificate this is done by generators and orbit enumeration.

def preserves_code(perm,C):
    image={tuple(c[perm[i]] for i in range(24)) for c in C}
    return image == set(C)

# group = [perm for perm in ambient_sextet_group if preserves_code(perm,G24)]
# assert len(group) == 9216

# Stage 6: orbitals on X x X
# pair_to_orb[x,y] is obtained by orbit enumeration under the action of group.
# orbitals = orbits_of_group_on_ordered_pairs(group,X)
# assert len(orbitals) == 985

# Stage 7: tensor by two-step incidence
# for each orbital gamma choose representative (x,y)
# count intermediate z according to the colors of (x,z) and (z,y)

# triples=[]
# for gamma,(x,y) in enumerate(reps):
#     C={}
i#     for z in range(len(X)):
i#         alpha=pair_to_orb[x,z]
i#         beta=pair_to_orb[z,y]
i#         C[(alpha,beta)]=C.get((alpha,beta),0)+1
i#     for (alpha,beta),n in C.items():
i#         triples.append((alpha,beta,gamma,n))
i# assert len(triples)==1414965

# Stage 8: center and blocks
# Solve z A_i = A_i z for all orbital matrices A_i.
# Split a generic central element over the checked primes.
# Recover block sizes from dim(Ae)=m^2.

# Stage 9: automorphisms
# Build a separating fingerprint base and enumerate all compatible base images.
# The count is 9216, hence the full automorphism group is the acting group.

```

The commented stages are expanded in the verification scripts because a direct unoptimized enumeration of the ambient group and tensor is slow. The mathematical content is unchanged: every stored datum is obtained by orbit enumeration, incidence counting, linear algebra over finite fields, or base-image search.

B Independent consistency checks

The finite object has several checks that do not depend on one another.

B.1 The square-sum check

The algebra has dimension 985. The Wedderburn sizes satisfy

$$\begin{aligned} & 7 \cdot 1^2 + 8 \cdot 2^2 + 4 \cdot 3^2 + 8 \cdot 4^2 + 4 \cdot 5^2 + 2 \cdot 6^2 + 8^2 + 9^2 + 2 \cdot 10^2 + 11^2 + 12^2 \\ &= 7 + 32 + 36 + 128 + 100 + 72 + 64 + 81 + 200 + 121 + 144 \\ &= 985. \end{aligned}$$

Thus the block list exhausts the full orbital algebra.

B.2 The quotient square-sum check

The 236-quotient has block sizes

$$1^{20}, 3^{10}, 4, 5, 6, 7.$$

They satisfy

$$20 \cdot 1^2 + 10 \cdot 3^2 + 4^2 + 5^2 + 6^2 + 7^2 = 20 + 90 + 16 + 25 + 36 + 49 = 236.$$

Thus the first quotient is also fully accounted for as a semisimple algebra.

B.3 The object-matrix check

The six-object matrix has entries summing to 985. Its diagonal entries sum to

$$33 + 16 + 14 + 86 + 44 + 104 = 297.$$

This is the number of diagonal orbital unknowns used in the center calculation. Thus the block-level diagonal count agrees with the center-system construction.

B.4 The shell check

The Golay enumerator gives 2576 dodecads. The sextet profile split gives

$$576 + 720 + 1280 = 2576.$$

The lifted coorient split gives

$$384 + 192 + 144 + 576 + 512 + 768 = 2576.$$

Both decompositions independently exhaust the same shell.

B.5 The automorphism check

The group used to build the coherent configuration has order 9216. The base-fingerprint search finds exactly 9216 color-compatible base images. Since every acting group element gives one compatible base image and every compatible base image extends to a unique automorphism, the color-preserving automorphism group has exactly the same order as the acting group.

C Certificate A: the root-killing kernel

The following kernel is the complete root-killing calculation. It constructs H_8 , forms $H_8^{\oplus 3}$, applies the three neighbors, and prints the root counts.

```
from itertools import product

def add(u,v): return tuple(a^b for a,b in zip(u,v))
def dot(u,v): return sum(a*b for a,b in zip(u,v)) % 2
def wt(u): return sum(u)

def span(gens):
    S={0,)*len(gens[0])}
    for g in gens:
        S |= {add(x,g) for x in list(S)}
    return sorted(S)

pts=list(product([0,1], repeat=3))
gens=[tuple(1 for _ in pts)]
for j in range(3):
    gens.append(tuple(p[j] for p in pts))
H8=span(gens)
C=[a+b+c for a in H8 for b in H8 for c in H8]

def vec(S): return tuple(1 if i+1 in S else 0 for i in range(24))
def neighbor(C,v): return span([c for c in C if dot(c,v)==0]+[v])
def roots(C): return sum(1 for c in C if wt(c)==4)

v1=vec({2,4,9,10,12,13,14,16,17,18,20,22})
v2=vec({3,4,9,13,15,16,19,20})
v3=vec({2,5,7,8,9,10,12,14,15,16,17,18,19,21,22,23})
for v in [v1,v2,v3]:
    print(roots(C))
    C=neighbor(C,v)
print(roots(C))
```

The output is 42, 18, 6, 0.

D Certificate B: orbital tensor

Given the pair-color array `pair_to_orb` and one representative pair for each orbital, the multiplication tensor is exactly this count:

```
triples=[]
for gamma,(x,y) in enumerate(representative_pairs):
    C={}
    for z in X:
        a=pair_to_orb[x,z]
        b=pair_to_orb[z,y]
        C[(a,b)]=C.get((a,b),0)+1
    for (a,b),n in C.items():
        triples.append((a,b,gamma,n))
```

The number of stored nonzero triples is 1,414,965. This is not a sampled count; it is the full tensor support.

E Certificate C: center and blocks

The center calculation solves the linear system

$$\sum_i x_i A_i A_j = \sum_i x_i A_j A_i, \quad j = 0, \dots, 984.$$

The nullity is 39. A generic central element splits into 39 primitive central idempotents over the checked primes. For each idempotent e the dimension of Ae is a square; the square root is the block size. The recovered multiset is

$$1^7, 2^8, 3^4, 4^8, 5^4, 6^2, 8, 9, 10^2, 11, 12.$$

F Certificate D: automorphisms

The automorphism certificate is based on an injective fingerprint. The base $B = (b_1, \dots, b_r)$ is chosen so that

$$x \mapsto (\text{orb}(b_1, x), \dots, \text{orb}(b_r, x), \text{orb}(x, b_1), \dots, \text{orb}(x, b_r))$$

separates all 2576 dodecads. A color-preserving automorphism is determined by the image of this base. Exhaustive search through compatible base images yields exactly 9216 possibilities, all realized by the acting group.

G Certificate E: files and hashes

The reproducibility data attached to this paper are listed below. The verification bundle stores the complete manifest; shortened hashes are printed here.

- `Gnatural_certification_report.json`: 11937 bytes, SHA-256 0af01ffb2673...8721e0d1c5a5.
- `golay_fast_certificate_report.json`: 2453 bytes, SHA-256 8ef0ab3687a3...171a4a1e0e44.
- `golay_fast_certificate.py`: 12564 bytes, SHA-256 af990640de05...20dcfe4479b2.
- `golay_wedderburn_report.json`: 18203 bytes, SHA-256 f77345424859...a301fef09d35.
- `golay_wedderburn_certificate_v3.py`: 28321 bytes, SHA-256 80a76603bd4b...f83a11f3039c.
- `Gnatural_automorphism_certificate.json`: 2684 bytes, SHA-256 4a1b4ebfbf04...79344bed0cf9.
- `gnatural_automorphism_full_certificate.py`: 2409 bytes, SHA-256 8ff1a16dad47...af37abc5e930.
- `Gnatural_all_simple_matrix_model_atlas.json`: 61548 bytes, SHA-256 3482f57ac4aa...d9c0b1d6c9c2.
- `Gnatural_all_simple_matrix_models.npz`: 61477 bytes, SHA-256 e96e5f4c3507...ff81ede4f99f.
- `golay_985_to_236_branching.json`: 67355 bytes, SHA-256 6561b582f27e...d87c682ee4ba.
- `golay_985_to_42_12_branching_by_dvectors.json`: 35805 bytes, SHA-256 ff32f96b92e8...c9f0e56d0fd5.
- `golay_A42_T6_to_A12_restriction.json`: 3109 bytes, SHA-256 468f822a3b9c...dfb97dd6e7a0.
- `golay_packet20_profile.json`: 14134 bytes, SHA-256 0a65c50db789...b74dce1b9634.
- `golay_packet20_matrix_model.json`: 7336 bytes, SHA-256 18c5cc691b91...13423a95c78a.
- `golay_full_tensor_out/golay_985_tensor_sparse.npz`: 5176071 bytes, SHA-256 5d638460422d...d10a170.
- `golay_full_tensor_out/golay_985_orbital_reps.json`: 66890 bytes, SHA-256 ece64362c11d...44bbdd2e0.
- `Gnatural_final_certificate_bundle.zip`: 4985069 bytes, SHA-256 9cfd1337d935...c401e9d3831d.

H Notation

H_8 The extended Hamming code RM(1,3).

C_0 The direct sum $H_8^{\oplus 3}$ in $\mathbb{F}_2^{24}$.

$N_v(C)$ The Type II neighbor $\langle C \cap v^\perp, v \rangle$.

G_{24} The extended binary Golay code obtained after the three neighbor moves.

X The dodecad shell $G_{24}^{(12)}$.

$\tilde{B}_3$ The code-preserving lifted coorient group of order 9216.

R_α An orbital of $\tilde{B}_3$ on $X \times X$.

A_{985} The coherent algebra spanned by the 985 orbital relation matrices.

$G^\natural$ The same algebra A_{985} , regarded as the natural object.

End of proof

The construction begins from the forty-two roots of $H_8^{\oplus 3}$, removes them by three explicit Type II neighbors, extracts the dodecad shell of the Golay endpoint, closes the lifted coorient action on ordered pairs, and obtains a semisimple coherent algebra whose full automorphism group is the group that built it. This proves the finite closure of $G^\natural$.